

Sai Muneesh Puligundla

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Summary

Applied AI Engineer with experience spanning enterprise software, language systems, and machine learning model development. At Accenture, worked on backend services and Python based ML workflows; at Wright State University, developed LLM and NLP systems; at Humanitarian Technology Labs, built computer vision and healthcare ML solutions. Brings a strong software foundation and a practical approach to building reliable AI systems.

Technical Skills

Programming & ML: Python, PyTorch, TensorFlow, Scikit-learn, Apache Spark, NLP, Computer Vision, Feature Engineering, Model Evaluation

LLM & Retrieval: OpenAI API, LangChain, LangGraph, Hugging Face Transformers, LLMs, RAG, Prompt Engineering, Fine-Tuning (LoRA/PEFT), AI Agents

Backend & Data: FastAPI, Flask, Spring Boot, REST APIs, Microservices, PostgreSQL, MySQL, MongoDB, ETL Pipelines, FAISS, Pinecone

Cloud & MLOps: Azure, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Terraform, MLflow, Experiment Tracking

Enterprise & Tools: Microsoft Dynamics 365, Azure ML, Azure Data Factory, Power Platform, Power BI, Datadog, Grafana, Prometheus, Git, Linux

Professional Experience

Wright State University

Artificial Intelligence Research Intern

Jan. 2025 — Apr. 2026

- Designed and evaluated multi-agent LLM architectures for document intelligence and knowledge synthesis, applying CRAG and SELF-RAG retrieval patterns to address core failure modes in standard RAG pipelines, specifically hallucination under low-relevance retrieval and query answer misalignment
- Built structured evaluation frameworks to measure LLM system behavior across factual grounding, reasoning coherence, and task completion; used findings to drive iterative improvements to retrieval strategy, context window management, and prompt design
- Investigated agentic orchestration using LangGraph, including supervisor worker topologies, persistent state graph design for multi-step reasoning, and tool use coordination across agents handling heterogeneous data sources
- Benchmarked prompt engineering strategies (chain-of-thought, structured output via Pydantic, few shot with dynamic example selection) across OpenAI and open source models on domain specific research tasks
- Built reproducible ML research infrastructure covering experiment tracking, versioned datasets, and modular Python codebases that integrated cleanly with the lab's existing tooling and enabled other researchers to build on the work directly

Accenture

Software Engineer — Advanced App Engineering Analyst

Jul. 2022 — Dec. 2023

- Built enterprise workflows and backend services for Microsoft Dynamics 365 CRM and ERP, integrating business logic, operational data, and Azure based services for client platforms
- Developed Python based ML workflows for lead scoring, opportunity prioritization, and demand forecasting using CRM activity, transaction history, and supply chain data
- Designed data pipelines with Azure Data Factory and SQL based systems to prepare training data, support scheduled retraining, and improve consistency across reporting and model inputs
- Built and integrated REST APIs and microservices using Spring Boot and Python to connect business applications, analytics services, and downstream data workflows
- Managed model deployment and operational monitoring through Azure ML, Azure Application Insights, and Grafana, helping teams track prediction quality and service reliability
- Improved system responsiveness and stability through query tuning, logging, observability, and resilient integration patterns across distributed services

Humanitarian Technology Labs (HuT Labs), Amrita University

Research Engineer — AI/ML Systems

Jan. 2019 — Apr. 2022

- Designed sensor data acquisition systems on Arduino Mega, integrating ECG (AD8232), IMU (MPU-6050), and environmental sensor modules with custom shield layouts; wrote firmware to handle sampling rate configuration, analog noise filtering, and reliable serial streaming to a Python host
- Built the complete ECG anomaly detection pipeline: analog signal capture via AD8232 and Arduino Uno, digital bandpass filtering and R-peak detection in Python, HRV feature extraction, and a CNN-LSTM classifier trained on labeled cardiac data — deployed as a real-time inference service on Raspberry Pi 4, with regulated power delivery through a dedicated SMPS module to keep analog front-end noise below measurement thresholds
- Trained YOLO and SSD object detection models in PyTorch on a 10,000+ image dataset; exported to ONNX and optimized with TensorRT for deployment on NVIDIA Jetson Nano, achieving frame rates suitable for real-time robotic operation in constrained field conditions

- Managed SMPS-based power distribution for multi-board embedded systems, designing isolated 5V and 3.3V supply rails to protect noise-sensitive analog sensor circuitry from switching transients introduced by digital and compute hardware
- Built IoT autonomous control systems using Arduino as the sensor and actuation layer, communicating over I2C and UART with a Raspberry Pi running Python decision logic — fully offline, with no cloud dependency and sub-second response loops
- Handled the hardware-to-model lifecycle end to end: sensor calibration, raw signal validation, feature engineering, cross-validation, and deployment onto constrained hardware — iterating on both firmware behavior and model architecture in parallel to meet simultaneous latency and accuracy requirements
- Published peer-reviewed IEEE research on ML for cardiovascular disease prediction, accepted at IEEE ASIANCON 2022

Technical Projects

Multi-Agent Research Intelligence System *Python, LangGraph, LangChain, OpenAI API, Tavily, FastAPI, FAISS, PostgreSQL, React, Docker, AWS ECS*

- Architected a supervisor-worker multi-agent system using LangGraph where specialized agents handle web search (Tavily API), document retrieval, claim verification, and structured report synthesis — the supervisor dynamically routes tasks based on intermediate agent findings rather than executing a fixed pipeline
- Implemented CRAG-style retrieval evaluation inside the search agent: retrieved documents are scored for query relevance before passing downstream; below-threshold results trigger targeted web search augmentation, keeping the final context grounded without over-fetching
- Built a persistent shared memory state graph across agent hops using LangGraph's state management, enabling coherent multi-step reasoning over long research tasks without context bleed between unrelated branches
- Deployed as a FastAPI service on AWS ECS with a React frontend; FAISS and PostgreSQL back the retrieval layer, with Prometheus metrics tracking agent hop count, retrieval quality distribution, and end-to-end query latency

CRM Intelligence & Automation Engine *Python, LangChain, OpenAI API, Dynamics 365 API, Azure ML, Azure Functions, BERT, PostgreSQL, Power BI, Docker*

- Built an LLM-powered intelligence layer over Dynamics 365 CRM that classifies incoming support tickets, generates contextual response drafts using RAG over a historical case knowledge base, and scores active accounts for churn risk using behavioral signals from CRM activity data
- Fine-tuned a BERT classifier on labeled CRM ticket data for intent and category detection; integrated inference into D365 via Azure Functions triggered on case creation events — auto-routes tickets and pre-fills case fields before an agent opens the record
- Designed a RAG pipeline backed by a PostgreSQL store of resolved support history — response drafts are generated with OpenAI GPT conditioned on retrieved similar cases, targeting high-repetition ticket categories where manual drafting created bottlenecks
- Exposed churn risk scores and ticket analytics through a Power BI dashboard connected to the Azure ML model registry, giving account teams real-time visibility into customer health signals across their portfolios

Certifications

- Microsoft Engineer Program (In Progress) — AI Engineer (2025–2026)
- Microsoft Certified: Azure Fundamentals (AZ-900) — Jun 2023
- Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900) — Aug 2023
- Microsoft Certified: Power Platform Fundamentals (PL-900) — Nov 2023
- Microsoft Certified: Dynamics 365 Fundamentals, CRM (MB-910) — 2023
- Machine Learning Certification — *[Update issuer: Coursera / deeplearning.ai / other]*
- Accenture Certified Full Stack Developer — 2022

Publications & Leadership

IEEE Publication: *Developing Machine Learning Models for Cardiovascular Disease Prediction* — IEEE ASIANCON 2022 — DOI: 10.1109/ASIANCON55314.2022.9908772

IEEE Executive Committee Member (2019–2021): Organized technical workshops and mentored 20+ junior students on AI, robotics, and embedded systems engineering

Education

Wright State University

Master of Science in Computer Science — GPA: 3.8/4.0

Dayton, OH

Jan. 2024 — May 2026

- Coursework: Machine Learning, Artificial Intelligence, Distributed Computing, Algorithm Design & Analysis, Parallel Programming, Information Retrieval

Amrita Vishwa Vidyapeetham University

Bachelor of Technology in Electronics and Communication Engineering

Jun. 2018 — Apr. 2022